

Instruction Robustness in RAG Pipelines

Isolating the System Prompt as a Causal Variable

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Outline

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The problem

RAG in one line. Retrieve evidence from an external knowledge base, then let a language model generate an answer from it (Lewis et al., 2020).

Three things shape the answer:

- the query the user sends,
- the evidence the retriever supplies,
- **the system prompt** — the standing instruction the engineer writes.

Why the system prompt is special

It is the only one of the three that is *deliberately authored*.

⇒ If the prompt is the part we author, we should know what authoring it actually does.

Research question

RQ

To what extent do systematic changes in the system prompt alter the behaviour and outcomes of a RAG system when the user input, the knowledge base, the retrieval mechanism, and the underlying language model are held constant?

Answering it needs exactly two things, and the thesis builds both:

- 1 a **formal model** of the system precise enough to isolate the system prompt as a single parameter (Ch. 4);
- 2 an **experimental design** in which every other source of variation is held fixed (Ch. 5).

Scope. Standard single-shot RAG — deliberately.

The system as a chain

$$(q, D) \longrightarrow p_{\eta}(z \mid q, D) \longrightarrow p_{\theta}(y_i \mid q, z, y_{1:i-1}, S) \longrightarrow p_{\text{RAG}}(y \mid q, D, S)$$

q user query

S system prompt (standing instruction)

$D = \{d_1, \dots, d_N\}$ knowledge base

z a single retrieved document

$y = (y_1, \dots, y_L)$ generated answer

$C = \{z_1, \dots, z_K\}$ retrieved context

η, θ retriever / generator parameters

⇒ The system prompt S enters at the *generation* stage only. The research question is then a question about one distribution: what changes in $p_{\text{RAG}}(y \mid q, D, S)$ when S changes?

Step 1 — probabilistic retrieval

Retrieval is a *distribution* over documents:

$$p_{\eta}(z \mid q, D)$$

The retriever never commits to one document being correct; it spreads probability over candidates.

Evaluating all N documents is impractical, so keep the top K :

$$\mathcal{Z}_K(q, D) = \underset{z \in D}{\text{TopK}} \ p_{\eta}(z \mid q, D), \quad C = \{z_1, \dots, z_K\}, \quad z_k \in \mathcal{Z}_K(q, D)$$

⇒ z is a candidate the model *sums over*; C is the concrete collection the running system retrieves and logs.

Step 2 — prompt-conditioned generation

Lewis et al. (2020) write the generator at token level as $p_{\theta}(y_i \mid q, z, y_{1:i-1})$.

The one extension this thesis makes

$$p_{\theta}(y_i \mid q, z, y_{1:i-1}, S)$$

The departure is deliberately small: *same model, same parameters* θ . What changes is that the system prompt is written into the conditioning information instead of being left implicit.

Generation is autoregressive, so a full sequence follows by multiplication:

$$p_{\theta}(y \mid q, z, S) = \prod_{i=1}^L p_{\theta}(y_i \mid q, z, y_{1:i-1}, S)$$

⇒ Making S an explicit argument is what turns "prompt engineering" into a variable that can be intervened on.

Step 3 — marginalising over retrieved documents

The system does not know which document supports the answer, so it averages over candidates (Lewis et al., 2020).

RAG-Sequence — one document explains the whole answer

$$p_{\text{RAG-Sequence}}(y \mid q, D, S) \approx \sum_{z \in \mathcal{Z}_K(q, D)} p_{\eta}(z \mid q, D) \prod_{i=1}^L p_{\theta}(y_i \mid q, z, y_{1:i-1}, S)$$

RAG-Token — different parts may come from different documents

$$p_{\text{RAG-Token}}(y \mid q, D, S) \approx \prod_{i=1}^L \sum_{z \in \mathcal{Z}_K(q, D)} p_{\eta}(z \mid q, D) p_{\theta}(y_i \mid q, z, y_{1:i-1}, S)$$

The whole difference is the order of the two operators: $\sum_z \prod_i$ against $\prod_i \sum_z$.

⇒ The choice does not matter here: S appears in *exactly the same place* in both — inside the generator and nowhere else. Both are written $p_{\text{RAG}}(y \mid q, D, S)$.

Decomposing the prompt: what it asks vs. how it asks

S enters the model as one symbol, but it is a text with internal structure — and that structure is what the experiment varies.

Requirements and wording

$$S = \rho(\Sigma, \varphi), \quad \Sigma = \{\sigma_1, \dots, \sigma_m\}$$

- Σ : the set of **requirements** the prompt states. Each σ_k is a property an answer either has or does not have, so $\sigma_k(y) \in \{0, 1\}$ is *checkable on any output* (Zhou et al., 2023).
- φ : the **wording** those requirements are expressed in.

Two prompts built from the same Σ ask for the same thing; two built from the same φ say things the same way.

⇒ This separation is the whole experimental lever: hold Σ fixed, vary φ , and any behavioural difference is an effect of wording alone.

The design: three requirement sets × five wordings

Factor 1 — requirement set (three published prompts, see Sec. 8):

Set	Source	Requirements m
Σ_A	response-generation prompt, Friel et al. (2024)	1
Σ_C	comprehensive instruction, Mao et al. (2024)	3
Σ_B	VANILLA instruction, Gao et al. (2023)	≈ 10

Factor 2 — wording, per baseline $b \in \{A, B, C\}$:

Condition	Type	What changes	Axis follows
$S_{b,0} = \rho(\Sigma_b, \varphi_0)$	canonical	nothing — the published form; the reference	—
$S_{b,1} = \rho(\Sigma_b, \varphi_1)$	lexical	synonym substitution, order and voice fixed	Zhan et al. (2024)
$S_{b,2} = \rho(\Sigma_b, \varphi_2)$	syntactic	clause reordering, active $\rightarrow$ passive	Kostić et al. (2026)
$S_{b,3} = \rho(\Sigma_b, \varphi_3)$	format	same wording as a list rather than prose	Sclar et al. (2024)
$S_{b,4} = \rho(\Sigma'_b, \varphi_0)$	deontic	Σ itself changes: obligation $\rightarrow$ request	—

$\Rightarrow 3 \times 5 = 15$ prompt conditions; every contrast taken *within* a baseline. **Neither factor is invented here:** the requirement sets are published RAG prompts, and the wording axes are the linguistically defined, meaning-preserving paraphrase types of Wahle et al. (2024). What is new is applying them to the *standing system prompt* of a retrieval-augmented system.

Concrete: the five conditions of baseline A

A0	Use the following pieces of context to answer the question.	<- canonical
A1	Employ the following excerpts of material to address the query.	<- lexical
A2	The question is to be answered using the context pieces below.	<- syntactic
A3	CONTEXT: the following pieces TASK: use them to answer the question	<- format
A4	Please try to answer the question using the context pieces below.	<- deontic

Within a baseline all five stay within $\approx 10\%$ of the canonical length, so a difference cannot be read as an effect of prompt length.

Why three baselines and not one

The empirical question

If the system responded only to *what* a prompt requires and not to *how* it is written, then $S_{b,0}$ through $S_{b,3}$ would produce the same output distribution within every baseline — they differ in φ alone. **Whether they do is the experiment.**

The causal structure (Figure 4.1)

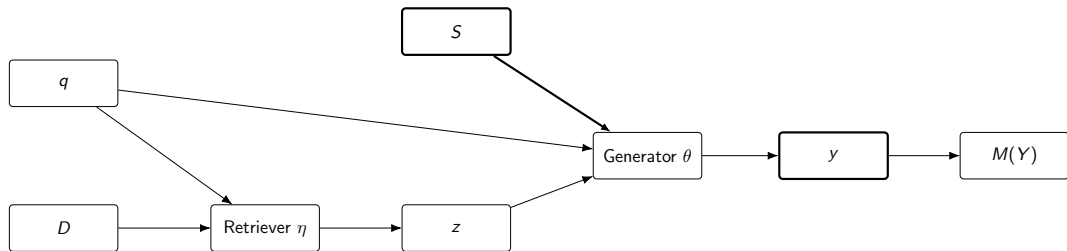


Figure 1: The system prompt S and the output y are drawn in bold. S enters the generator and has **no arrow into the retriever**, which is what allows a change in y to be attributed to a change in S .

Read off the graph: $S \rightarrow y$ exists; $S \rightarrow z$ does not. In the notation of Pearl (2009), S has an effect on y and no effect on z .

The missing edge, and what enforces it

Held constant across conditions: each query fixed (paired design), $D = D_0$, $\eta = \eta_0$, $\theta = \theta_0$.

Decoding remains stochastic, and the model says so explicitly:

$$Y = G(q, z, S; \theta, \varepsilon)$$

$\Rightarrow \varepsilon$ is why every condition is *repeated* ($n = 5$) rather than run once.

From intervention to estimand

Setting the prompt is an **intervention**, $\text{do}(S = S_j)$ (Pearl, 2009).

- *Setting* the prompt while holding the rest fixed does not.

For one query: $Y \sim p_{\text{RAG}}(y \mid q, D, S_j)$. But a single query is not the quantity of interest — a prompt that helps one question and hurts the next has established no effect at all.

Average causal effect of condition S_j vs. baseline S_0

$$\tau_j = \mathbb{E}_{q \sim P_Q} \left[\mathbb{E}_{Y \sim p_{\text{RAG}}(\cdot \mid q, D, S_j)}[M(Y)] - \mathbb{E}_{Y \sim p_{\text{RAG}}(\cdot \mid q, D, S_0)}[M(Y)] \right]$$

⇒ This is an average treatment effect in the standard sense: the *query* is the unit, the *prompt condition* the treatment.

What would falsify the hypothesis

Null

Within each baseline b , conditions $S_{b,0}, \dots, S_{b,3}$ share Σ_b and differ in φ alone. If the system responds only to *what* is asked, then $\tau_1 = \tau_2 = \tau_3 = 0$ for every b and every evidence level.

And the result is informative either way:

- $\tau_j \neq 0 \Rightarrow$ meaning-preserving rewording of a *system* prompt changes RAG behaviour — a robustness problem for every deployment that reuses a pipeline and edits only the prompt.
- $\tau_j \approx 0$ across all three baselines $\Rightarrow$ a non-trivial *negative* result: prompt-sensitivity findings from the non-retrieval setting (Salinas and Morstatter, 2024; Sclar et al., 2024) do *not* transfer to the standing system prompt in RAG.

Design: one independent variable, one robustness factor

Independent variable: the system prompt — 15 conditions.

Held at a fixed value for the whole of data collection:

Component	Setting
Generator	one instruction-tuned LLM; identity, θ , temperature, max length fixed
Embedding	one model, both sides, dense dual encoder (Karpukhin et al., 2020)
Index	FAISS (Johnson et al., 2021), built once, never rebuilt; exhaustive
Chunking χ	one passage per chunk
Depth	$K = 0, 5, 10$

Controlled robustness factor: evidence quality at four levels. A prompt effect that appeared only on clean evidence would be of little practical interest.

$$y_{ijkr} \sim p_{\text{RAG}}(y \mid q_i, \text{do}(C = c_{ik}), \text{do}(S = S_j)), \quad r = 1, \dots, n$$

Knowledge base

The corpus is collected, not generated. MS MARCO passage collection (Nguyen et al., 2016).

Contents	every passage attached to a validation row with a non-empty <code>wellFormedAnswers</code>
Size	$N = 122,678$ passages
Distractors	<i>none sampled</i> — gold passages compete against the whole corpus
Index	exhaustive, not approximate (0.38 GB of float32 vectors fits in memory)

Why an established dataset rather than an invented one?

- On an invented corpus the data model and the hypothesis are written by the same hand.

Query set: $N_Q = 100$

Two categories, fifty each, differing in *one* property: how far the evidence is spread across passages.

Q_{direct} — 50 queries

Answer carried by a *single* passage marked relevant. One passage suffices, so a failure is a failure to use evidence that was present. The noise-robustness testbed of Chen et al. (2024).

$Q_{\text{multi-hop}}$ — 50 queries

Human-written answer draws on *two or more* relevant passages; the annotation identifies which. The requirement to combine evidence is externally recorded, not asserted. The information-integration testbed.

What counts as a good answer

Requirement	Measured by	Source
Grounded	faithfulness	Es et al. (2023), reference-free
Relevant	answer relevance	Es et al. (2023), reference-free
Correct	accuracy vs. annotation	Chen et al. (2024)
Obedient	instruction adherence	prompt-level strict accuracy, Zhou et al. (2023)

The first three are the measures Yu et al. (2024) name as the evaluation targets of the *generation* component — each a pairwise relation: relevance (response↔query), faithfulness (response↔documents), correctness (response↔reference).

Adherence is checked per baseline — and the judge is audited

No common checklist

A condition is held only to what *its own* prompt states.

Σ_B fixes citation syntax and density $\Rightarrow$ parse document identifiers, count per sentence.

- Σ_C requires a justification $\Rightarrow$ look for a justification marker.
- Σ_A states nothing beyond grounding $\Rightarrow$ its adherence measure is *undefined by construction*, not omitted; faithfulness carries the whole of what it asks.

A single shared checklist would score conditions against requirements their prompts never stated.

The judge

Correctness and adherence are computed **by program**. Faithfulness and answer relevance are scored by a separate LLM (Es et al., 2023; Saad-Falcon et al., 2024).

The prompts are not composed from scratch (1/2)

Their wording is taken from prompts **published with retrieval-augmented systems**, so the baseline is a form already in use rather than one written to be improved upon.

Σ_A — **the fewest requirements**. Response-generation prompt of Friel et al. (2024): grounds the answer in the documents, asks the question, does nothing else.

```
Use the following pieces of context to answer the question.  
{documents}  
Question: {question}
```

Σ_B — **the most**. VANILLA instruction of Gao et al. (2023) (their Table 23). Beyond grounding it constrains accuracy, engagement, concision, register, and citation syntax and density.

```
Write an accurate, engaging, and concise answer for the given question using  
only the provided search results (some of which might be irrelevant) and cite  
them properly. Use an unbiased and journalistic tone. Always cite for any  
factual claim. When citing several search results, use [1][2][3]. Cite at  
least one document and at most three documents in each sentence. If multiple  
documents support the sentence, only cite a minimum sufficient subset.
```

The prompts are not composed from scratch (2/2)

Σ_C — in between, and the direct precedent for $\rho(\Sigma, \varphi)$. Mao et al. (2024) describe their instruction as consisting of *three parts*, with a stated rationale per part: direct the model to the passages; ask it to read the question carefully; ask for an answer *plus* a justification.

```
Refer to the passage below and answer the following question.  
Make sure you fully understand the meaning of the question and passages.  
Then give the answer and explain why you choose this answer.  
Passages: {passages}  
Question: {question}
```

- Writing a prompt as an *enumerated set of requirements with a purpose per requirement* is exactly what $\rho(\Sigma, \varphi)$ formalises. Mao et al. (2024) report this form outperforms both a bare instruction and a chain-of-thought variant on the same passages.
- **No requirement checked here is an invention of this thesis:** the citation constraint is the central requirement of Gao et al. (2023); the justification constraint is the third part of Mao et al. (2024). That is why adherence can be checked by *parsing the output* rather than by judgement.

Limitations, stated up front

- **A single collection.** Every result is conditional on MS MARCO — its queries, passages, relevance judgements and written answers. Topical spread *within* one corpus is not replication *across* several. A prompt effect established here bounds what may be expected elsewhere rather than establishing it.
- **Composition of that collection.** MS MARCO records what people type into a web search engine: short, largely factoid, web-domain queries. Clinical decision support and legal QA — two of the deployments that motivate RAG in the first place — are absent, and their documents differ in length, structure and register.
- **No closed world.** Φ holds the annotated answers and the facts in D , a subset of everything true. Correctness is therefore evaluated only over claims bearing on a_q ; a true fact outside Φ is not counted against the answer.
- **Single-shot architecture only.** By design — see the scope argument. Graph-structured and agentic pipelines are the natural extension, and this is the controlled baseline they need.

Summary

- 1 **Formal model.** The RAG formulation of Lewis et al. (2020), extended by one variable: the system prompt becomes an explicit argument of the generator, $p_{\theta}(y_i \mid q, z, y_{1:i-1}, S)$, and therefore of $p_{\text{RAG}}(y \mid q, D, S)$.
- 2 **Prompt decomposition.** $S = \rho(\Sigma, \varphi)$ separates *what* a prompt requires from *how* it is worded — 15 conditions: 3 published requirement sets $\times$ 4 wordings + 1 deontic variant. Neither factor is invented: the sets are published RAG prompts, the axes the paraphrase types of Wahle et al. (2024), Zhan et al. (2024) and Kostić et al. (2026).
- 3 **Causal model.** S has an arrow into the generator and none into the retriever (Fig. 4.1); the missing edge is enforced by the two-stage procedure, not merely assumed. The estimand is τ_j , an average treatment effect over P_Q .
- 4 **Experiment.** 15×4 cells, 100 MS MARCO queries, $n = 5$ (the budget of Zhuo et al., 2024), 30,000 generations; four published generation-stage metrics; an audited LLM judge; a falsifiable null.
- 5 **Realization.** The assumption is enforced by the code, not asserted: retrieval and generation are separate passes across a disk barrier, every controlled parameter lives in one file, and the primary cost function has no free parameters.

$\Rightarrow$ Every other explanation for a difference between conditions is removed *before* data collection — which is what licenses reading $\hat{\tau}_j$ as the effect of the system prompt.

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Thank you — questions welcome.